

**Psych 507A: Statistical Methods in Psychological Research
Fall Semester 2014**

Lecture: Tuesdays & Thursdays, 11:00 AM - 12:15 PM, Psychology 205
Office Hours: Tuesdays & Thursdays, 9:30 AM - 11:00 AM, Psychology 440
Professor: Aurelio José Figueredo, PhD; Telephone: 621-7444
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Textbooks:

- **REQUIRED (QASS BOOKS):**
 - Understanding Regression Analysis (QASS #57), SAGE Publications (ISBN: 0803927584)
 - Applied Regression (QASS #22), SAGE Publications (ISBN: 0803914946)
 - Multiple Regression in Practice (QASS #50), SAGE Publications (ISBN: 0803920547)
 - Regression with Dummy Variables (QASS #93), SAGE Publications (ISBN: 0803951280)
- **RECOMMENDED (QASS BOOKS):**
 - Interpreting and Using Regression (QASS #29), SAGE Publications (ISBN: 0803919158)
 - Regression Diagnostics (QASS #79), SAGE Publications (ISBN: 080393971X)
 - Analysis of Variance (QASS #1), SAGE Publications (ISBN: 0803930011)
 - Analysis of Covariance (QASS #12), SAGE Publications (ISBN: 0803911645)
- **RECOMMENDED (NON-QASS BOOKS):**
 - Cohen, Cohen, West, & Aiken (2003). Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences, Third Edition. Lawrence Erlbaum (ISBN: 0805822232)
 - Keppel & Wickens (2004). Design and Analysis: A Researcher's Handbook. Prentice Hall (ISBN: 0135159415)
- In addition, copies of source articles will be made available for download on D2L.

Grading:

There will be one Midterm and one Final exam. Both tests will be essay exams. Initially, an average will be computed for the course, with the Midterm and Final examination being weighted equally. However, provided both tests have been completed on schedule, no one will receive a letter grade that is lower than that of the cumulative Final examination. The Final exam score can therefore enhance the grades of late or gradual improvers if it is higher than the preceding performance without disproportionately influencing the grades of early or consistent performers.

Emphasis:

The emphasis of this course will be on the relationship of current statistical modeling procedures to both research pragmatics and paradigms, as well as to dominant metaphysical assumptions regarding both the substantive contents and methodological processes of behavioral science. The goal will be to develop the requisite level of critical thinking and philosophical background for conducting valid quantitative research rather than a "cookbook" familiarity with traditional statistical procedures. This involves a conceptual understanding of the underlying philosophical basis of empirical science as well as of the mathematical foundations and interrelationships of the major statistical models now available. The readings will be used as a minimal framework for discussions of the major issues, but will not limit the maximal scope of the material covered. Class participation is therefore strongly encouraged.

Outline of Topics:

- Metascience
 - Philosophy of Science
 - Kant
 - Popper
 - History of Science
 - Kuhn
 - Feyerabend
 - Lakatos
 - Laudan
 - Methodology of Science
 - Chamberlin
 - Platt
 - Meehl
 - Shadish
 - Petrinovich
 - Sechrest
- Bivariate Correlation and Regression
 - Galton
 - Pearson
- Multiple Correlation and Regression
 - Multiple R-Squared
 - Model F-Ratio
 - Specification Error
 - Multicollinearity
 - Hierarchical Partitioning of Variance
 - Nested Model Comparisons
 - Non-Additivity (Interactions)
 - Non-Linearity (Polynomials)
- Exploratory Regression (Empirical Selection) Techniques
 - Backwards Elimination
 - Forward Selection
 - Stepwise Regression
 - Alpha Slippage
 - Shrunken R-Squared
 - Cross Validation
- General Linear Model/Regression Approach to ANOVA
 - Fisher
 - Classical ANOVA and its limitations
 - Omnibus Null Hypothesis
 - Dummy Coding
 - Contrast Coding
 - Planned Orthogonal Contrasts
 - Limitations of Post Hoc Comparisons
 - Mixed General Linear Models
 - Classical ANCOVA and its limitations
 - Modern Mixed GLMs
- Canonical Analysis
 - Simultaneous Canonical Analysis (SIMCA)
 - Sequential Canonical Analysis (SEQCA)
 - Other Variants of Canonical Analysis
 - Multivariate Analysis of Variance (MANOVA)
 - Multiple Discriminant Function Analysis (MDFA)
 - Pearson χ^2 Contingency Tables (CHISQ)

Relevant University Regulations:

1. Absence policies:
If the instructor is not in class within 20 minutes of the start of class, then class is cancelled.
 - a. We honor all holidays or special events observed by organized religions for those students who show affiliation with that particular religion.
 - b. We honor all absences pre-approved by the UA Dean of Students (or Dean's designee)
2. Policies about plagiarism:
Don't plagiarize. Find the Student Code of Academic Integrity:
<http://dos.web.arizona.edu/uapolicies/> and read it. Adhere to those rules.
3. Policies about threatening behavior by students:
Don't engage in threatening behavior. Find <http://policy.web.arizona.edu/~policy/threaten.shtml> and read it. Adhere to those rules.
4. Required extracurricular activities:
There are none.
5. Special materials required for the class:
There are none.
6. Notification:
Some of you may find the examples used in the main lecture (and expanded upon in this course) offensive. Sample data used in this course will often be based on theories/studies of human mating, genetics, reproductive success and failure, etc.
7. Students with Disabilities:
If you anticipate issues related to the format or requirements of this course, please meet with one of the instructors for this course. We would like to discuss ways to ensure your full participation in the course. If you determine that formal, disability-related accommodations are necessary, it is very important that you be registered with Disability Resources (621-3268; drc.arizona.edu) and that you notify us of your eligibility for reasonable accommodations. We can then plan how best to coordinate your accommodations.

Information contained in this syllabus, other than the grade and absence policies, may be subject to change with reasonable advance notice if we feel it is appropriate. Hence, the content of this syllabus is tentative.

GRADUATE STATISTICS SCREENING EXAMINATION

Please write a short paragraph doing a brief comparison and contrast of the pairs of terms given. Do not merely give definitions, but explain how the concepts are related to each other in terms of both similarity and difference.

1. Central Tendency vs. Dispersion
2. Mean vs. Median
3. Range vs. Variance
4. Continuous Variable vs. Categorical Variable
5. Descriptive Statistics vs. Inferential Statistics
6. Hypothesis Testing vs. Parameter Estimation
7. Null Hypothesis vs. Alternative Hypothesis
8. Type I Error vs. Type II Error
9. Sample vs. Population
10. Random Sampling vs. Random Assignment
11. Controlled Experiment vs. Observational Study
12. Association vs. Causation